shown: competitors for food and space, fellow travellers through time in the environment of the species. The 0.5 (0.125, etc.) refers to the *additional* relatedness *over and above* the background population, whose relatedness approaches zero.

Genes in the Williams sense are things you can count as the generations go by, and it doesn't matter what their molecular nature is; it doesn't matter, for instance, that they are split up into a series of 'exons' (expressed) separated by mostly inert 'introns' (ignored by the translation machinery). Molecular genomics is a fascinating subject but it doesn't heavily impinge on the 'gene's eye view' of evolution which is the central theme of the book. To put the point another way, *The Selfish Gene* is quite likely a valid account of life on other planets even if the genes on those other planets have no connection with DNA. Nevertheless, there are ways in which the details of modern molecular genetics, the detailed study of DNA, can be gathered into the gene's eye fold and it turns out that they vindicate that view of life rather than casting doubt on it. I'll come on to this after what may seem like a radical change of subject, beginning with a specific question which obviously stands for any number of similar questions.

How closely related are you to Queen Elizabeth II? As it happens, I know I'm her 15th cousin twice removed. Our common ancestor is Richard Plantagenet, 3rd Duke of York (1412–1460). One of Richard's sons was King Edward IV, from whom Queen Elizabeth is descended. Another son was George, Duke of Clarence, from whom I am descended (allegedly drowned in a butt of Malmsey wine). You may not know it but you are very probably closer to the Queen than 15th cousin and so am I and so is the postman. There are so many different ways of being somebody's distant cousin, and we are all related to each other in many of those ways. I know that I am my wife's twelfth cousin twice removed (common ancestor George Hastings, 1st Earl of Huntingdon, 1488–1544). But it is highly probable that I am a closer cousin to her in various unknown ways (various pathways through our respective ancestries) and it is absolutely certain I am also her more distant cousin in many more ways. We all are. You and the Queen might simultaneously be ninth cousins six times removed, and twentieth cousins 4 times removed, and thirtieth cousins 8 times removed. All of us, regardless of where in the world we live, are not only cousins of each other. We are cousins in hundreds of different ways. This is just another way of saying we are all